Angular Sparsity Is an Atomic-Sector Property: The Abelian Residue of Multi-Center Bonding*

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The GeoVac program builds molecular Hamiltonians from atom-centered Coulomb-Sturmian blocks whose angular matrix elements are generated from quantum-number labels by Gaunt selection, with no spatial quadrature. This yields structurally sparse qubit Hamiltonians—334 Pauli terms for LiH at $Q = 30$, $O(Q^{2.5})$ scaling (Paper 14)—at the cost of a documented binding failure for second-row diatomics (the W1e wall, Papers 19/20).

We settle the question of whether that trade was necessary. Using exact rational two-center machinery (Mulliken–Ruedenberg auxiliary functions for overlaps; a hydrogenic eigen-trick for cross-center one-body terms, certified by an exact bra/ket identity holding entrywise), we census the *genuine* bare tensor (S, h, g) for a real fragment pair and find: **[MEASURED]** the Gaunt l-selection rule does not survive cross-center evaluation. Only the axial m-rule does. Electron-repulsion density inflates 13.8× over the builder at $n_{\max} = 2$, rising to 15.0× at $n_{\max} = 3$, and cross-center one-body entries measure 0.19–0.80 Ha—bond scale, not perturbative.

We give the mechanism **[INTERNAL THEOREM]**: m labels the *abelian* symmetry shared by both centers (rotation about the internuclear axis is one global one-parameter group, so m is a common label), while l labels irreducible representations of a rotation group attached to a single center. The intersection of the two centers' rotation groups is exactly the axial $U(1)$, whose character group is $\mathbb{Z}$ by Pontryagin duality. Abelian selection rules are additive charge conservation and survive; non-abelian Clebsch–Gordan selection requires the full sphere and does not. Surviving label-generated sparsity is therefore governed by the largest common abelian symmetry, which is computable from the molecular point group in advance.

Two re-readings follow. First, the 17.9× Pauli inflation previously attributed to Löwdin orthogonalization (v4.73.1) was never orthogonalization's price: it is within measurement of the genuine tensor's own density, so Löwdin was paying an honest bill. Second **[OBSERVATION]**, four independent attempts to relocate the non-orthogonality—into the operator, into a dual basis, into a high-symmetry embedding, into device-measured many-body overlaps—each moved the cost without reducing it, because the cost measures the physical asymmetry of a two-center problem. A fifth, measured here, is the sharpest: enabling equivalent-atom-swap tapering *reduces* the qubit saving (BeH_2 12 → 9, N_2 22 → 19, F_2 22 → 15) while inflating the Pauli count 2.3–3.6×, because the swap rotation merges per-block abelian gradings into joint stabilizers. A permutation identifies blocks, and identifying blocks destroys the per-block labels that generate the sparsity.

We close with a demonstration **[PANEL-VERIFIED]**: all-electron 12-electron NaH, the framework's hardest documented failure, binds at $R_{\text{eq}} = 3.736\,a_0$ (+4.8% vs experiment) with 91.1% of in-basis FCI binding from three fragment-native determinants, once the integrals are genuine end-to-end. This localizes W1e to the Hamiltonian specification rather than the correlation treatment, by construction.

I. INTRODUCTION: WHAT THE SPARSITY BOUGHT, AND WHAT IT COST

The GeoVac framework's computational proposition is structural sparsity. Angular matrix elements are generated from the quantum-number labels (n, l, m) by Gaunt selection rules and Wigner $3j$ symbols with no spatial integration, so the resulting second-quantized Hamiltonian is sparse by construction rather than by numerical thresholding. The downstream consequence is the qubit-resource result of Paper 14 [2]: composed LiH at 334 Pauli terms on $Q = 30$ qubits, $O(Q^{2.5})$ Pauli scaling against $O(Q^{3.9–4.3})$ for Gaussian baselines, and a 190× reduction against cc-pVDZ.

The framework's documented failure is chemical binding for second-row diatomics. Named W1e, it has resisted an unusually thorough sequence of attacks: six Phillips–Kleinman modifications, basis-level Schmidt orthogonalization, core-correlation estimates, basis enlargement to $n_{\max} = 4$, an explicit-core Hartree–Fock treatment with twelve electrons, and a DMRG cross-check that reproduced FCI bit-identically at finite sector dimension without producing an interior minimum (Papers 19/20; CHANGELOG v2.19.4–v3.87.0). The wall was localized to the Hamiltonian specification—the projection from the continuous Level-4 solver to second-quantized $(h_1, \text{eri}, e_{\text{core}})$ integrals—rather than to the treatment of correlation.

This paper answers the question that localization leaves open. The framework neglects the overlap between orbitals on different centers, treating a non-orthogonal basis as though it were orthonormal, and that neglect is

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what preserves the sparsity. Was there an alternative? Could one keep the genuine cross-center physics *and* the label-generated sparsity, by declining to transform the integral tensor at all?

The answer is no, and the reason is group-theoretic rather than technical.

II. THE EXACT-ARITHMETIC CENSUS

A. Machinery and certificate

Two-center overlaps at arbitrary (n, l, m, Z) are computed as exact rationals via the Mulliken–Ruedenberg auxiliary functions A_n/B_n [3, 4], evaluated in prolate spheroidal coordinates where the two-center problem separates [1]. Because the result is rational, vanishing is *decidable* rather than inferred from a numerical tolerance—a property no quadrature scheme can offer, and the reason this census can make a claim about exact zeros at all.

Cross-center one-body elements are obtained by extending the same route with $1/r_A$ and $1/r_B$ kernels, each of which cancels against the volume element $(\xi^2 - \eta^2)$, followed by the hydrogenic eigen-trick. The assembly carries its own certificate: the bra/ket identity

$$E_a S - Z_B I_B \;=\; E_b S - Z_A I_A \tag{1}$$

holds as an *exact rational equality* on every entry, which is a machinery check no fortuitous cancellation survives. Four independent quadrature cross-checks agree at $\sim 2 \times 10^{-16}$.

B. Result

The census configuration is $Z_A = 3$ at the origin, $Z_B = 1$ at $R = 3\,\hat{z}$, hydrogenic orbitals to $n_{\max} = 2$ on each center, $M = 10$ spatial orbitals. Table I compares the genuine bare tensor against the production builder's.

Provenance, stated precisely because the three rows differ. The S and h rows are *computed*: every entry is evaluated in exact rational arithmetic by the machinery of Sec. II A, so both the zeros and the non-zeros are decided rather than inferred. The g row is *counted*: it is a symmetry-rule census in the complex-m basis, because no two-center electron-repulsion engine exists in the framework—the m-rule and per-side Gaunt feasibility are applied combinatorially, and no two-center electron-repulsion integral was numerically evaluated. The g figures are therefore claims about how many entries the selection rules *permit*, not verifications that each permitted entry is nonzero. Supplying that engine is the natural follow-on to this paper.

TABLE I. Genuine versus builder sparsity, $Z_A = 3/Z_B = 1$ at $R = 3\,a_0$, $n_{\max} = 2$ per center ($M = 10$). "Genuine zeros are" records which selection rule accounts for the surviving zeros.

object	genuine	builder	dense	genuine zeros are
S	32	10	100	m-rule only
h	44	22	100	m-rule only
g	2,944	214	10,000	m-rule + same-center Gaunt

Subject to that distinction, the electron-repulsion tensor is 29.4% permitted-dense genuinely against 2.1% for the builder.

Numerical corroboration of the g row. The counting can be checked without the missing Neumann engine, by evaluating the same tensor with a McMurchie–Davidson Gaussian engine over fitted Slater shapes. On the census configuration this gives 2,976 of 10,000 entries above 10^{-10}, i.e. 29.8% dense against the 29.4% counted—agreement at the level the basis convention permits, since the numerical evaluation is in the real Cartesian basis and the count is in the complex-m basis, which differ by the $\pm m$ mixing. So the permitted entries are *generically nonzero*: the density is a real property of the tensor and not an artifact of over-permissive counting. The l-selection failure appears at bond scale in individual entries—for instance $(1s_A\, 2p_{z,B} \,|\, 2s_A\, 2s_A) = -0.218$, where the same-center analog vanishes identically.

This does not upgrade the row to MEASURED: Gaussian-fitted floats make these threshold decisions rather than decided zeros, so the exact statement still awaits the engine. What it removes is the possibility that the counted density was an overcount.

The m-rule, verified without reference to m. Theorem 1(i) has a basis-free consequence that can be tested directly on the tensor: since $1/r_{12}$ is rotationally invariant and both nuclei lie on the axis, a rotation about the internuclear axis maps the basis into itself and must leave the tensor fixed. It does, to 2.2×10^{-16}. Displacing the nuclei off that axis and repeating the identical check gives 1.3×10^{-1}—fifteen orders of magnitude larger. The symmetry is axial, and only axial. At $n_{\max} = 3$ ($M = 28$) the comparison is 114,280 nonzeros (18.6%) against 7,600 (1.2%): an inflation factor of 13.8× rising to 15.0×, i.e. *increasing* with basis size.

Two further measurements matter for interpretation. First, per-side Gaunt feasibility removes less than 6% of entries beyond what the m-rule already removes, and the three cross classes $(AA|AB) + (AB|AB) + (AB|BB)$ account for 80% of the genuine tensor. The residual l-selection is therefore confined to the same-center $(AA|AA)$ and $(BB|BB)$ blocks. Second, cross-center magnitudes are large: overlaps 0.08–0.52, one-body entries 0.19–0.80 Ha. For scale, the entire NaH bond is under 2 eV, so individual entries in the dense part of the tensor exceed the binding energy being computed. They are not droppable as small corrections—an independent quantification of the W1d/W1e diagnosis.

III. MECHANISM: THE ABELIAN RESIDUE

The census result is not an accident of the chosen configuration. It is forced, and the forcing is elementary once stated in the right language.

Theorem 1 (Angular selection survives molecularization only in the abelian sector). *Let two nuclei A, B lie on the z-axis, with atom-centered basis functions labeled (n, l, m) about their own center, and let $\hat{O}$ be invariant under the molecular symmetry group. Then:*

(i) The m-selection rule holds exactly across centers: $\langle ab|\hat{O}|cd\rangle = 0$ unless $m_a + m_b = m_c + m_d$.

(ii) No l-selection rule holds between basis functions on different centers.

(iii) The surviving label-generated sparsity is exactly that generated by the largest symmetry group common to both centers, which is the axial $U(1)$ and is abelian.

Proof sketch. For (i): rotation about the internuclear axis is a *single* one-parameter group acting on all of $\mathbb{R}^3$. Its generator L_z is independent of which point on the axis is taken as origin, so m is a global label shared by both centers rather than a per-center one. Invariance of $\hat{O}$ under this group gives additive conservation of m.

For (ii): l labels irreducible representations of the rotation group about a *specific* center. Rotations about A and rotations about B are distinct subgroups of the Euclidean group whose intersection is precisely the axial subgroup. A function that is an l-eigenfunction about B, re-expanded about A, has support on all l: the two-center re-expansion does not terminate. Hence no l-selection between cross-center pairs. (The same-center blocks retain Gaunt selection, consistent with the $< 6\%$ residual measured in Sec. II.)

For (iii): the intersection in (ii) is $SO(2) \cong U(1)$, abelian. Abelian compact groups have one-dimensional irreducibles labeled by characters, and by Pontryagin duality the character group of $U(1)$ is $\mathbb{Z}$—which is the integer label m. Selection rules from abelian symmetry are additive conservation laws on integer charges, and are preserved by any operation respecting the axis. Non-abelian selection rules are Clebsch–Gordan couplings requiring the full group; reducing the group destroys them. □

The theorem is independently corroborated by the framework's own production code, along a path that knows nothing of this analysis: the cross-center nuclear-attraction routine in `shibuya_wulfman` returns zero by an explicit early exit when $m_1 \neq m_2$, and imposes no condition on l whatsoever. The m-rule is hard-coded because it holds; the l-rule is absent because it does not.

Theorem 1 converts a measurement into a prediction, because the largest common abelian symmetry is readable from the molecular point group before any integral is computed.

Corollary 1 (Sparsity budget from the point group). *For a molecule with point group G, label-generated angular sparsity is bounded by the grading available from the abelian structure of G. Linear molecules ($C_{\infty v}$, $D_{\infty h}$) carry a continuous $U(1)$ and therefore a $\mathbb{Z}$ grading, worth a multiplicative factor. Bent molecules carry only a finite point group and therefore a finite grading.*

H_2O has point group C_{2v}, which is order four and *abelian*, isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$; all its irreducibles are one-dimensional, so Corollary 1 predicts a two-bit spatial grading—two qubits, not a multiplicative factor. We attempted to test this and could not, for a reason that is itself a structural finding.

Observation 1 (The composed builder carries no angular geometry). *`molecular_spec.py` contains no bond angle: `h2o_spec` is `hydride_spec(8)`, parameterized by nuclear charge and a* single *O–H distance distributed across five blocks (core, two bond pairs, two lone pairs), with `MolecularSpec.nuclei = None`. The composed* H_2O *Hamiltonian therefore does not represent C_{2v} at all, and the framework's reported* H_2O *geometry error is a purely radial coordinate with no angular degree of freedom. Corollary 1 is consequently* not well posed *on this builder in the bent case; supplying hand-built bent nuclei would test a symmetry the Hamiltonian does not encode.*

Prediction 1 (Conditional, for any angular build). *In a build that does carry angular geometry, a molecule whose point group is abelian of order 2^k admits a k-bit spatial grading and no more. This remains falsifiable, but not on the present builder.*

What *is* measurable on the present builder is the linear case, where geometry is angle-free and therefore unambiguous—and it returns a result we did not anticipate, reported as Observation 3 below.

The measured factor of 3–5 for the m-rule at the bases of Sec. II is the $\mathbb{Z}$-grading case of Corollary 1. It is exact and structural, and it is a constant factor—not the $O(Q^{2.5})$ scaling story the framework's qubit results rest on.

IV. TWO RE-READINGS

A. Löwdin was paying an honest bill

The corpus previously recorded a 17.9× Pauli inflation from retrofitting Löwdin orthogonalization onto the balanced LiH integrals (CHANGELOG v4.73.1), and a 14× inflation in the earlier heterogeneous-nested experiment. Both were read as the price of orthogonalization, and orthogonalization was accordingly treated as the sparsity-destroying move.

The census shows this reading was wrong in an instructive way. The genuine tensor's own inflation over the builder is $\sim 15\times$, within measurement of the Löwdin figure. Orthogonalization was not destroying sparsity that the untransformed tensor possessed; the untransformed tensor never possessed it. "Löwdin destroys sparsity" and "the molecular tensor is intrinsically l-dense cross-center" are the same fact observed twice, from two sides.

This matters because it removes an entire class of proposed remedies. Any scheme whose premise is "avoid the transformation and keep the sparsity" is answering a question that was mis-posed.

B. Cost conservation

Observation 2 (Relocating the metric does not reduce it). *Four independent attempts to move the non-orthogonality out of the way each succeeded in moving it and failed to reduce it:*

1. *into the* ***operator*** *(Löwdin $S^{-1/2}$): operator densifies* 17.9×*;*
2. *into a* ***dual basis*** *(biorthogonal encoding): densifies comparably, and by theorem rather than happenstance [5];*
3. *into a* ***high-symmetry embedding*** *(hyperspherical Level 4N): angular basis requirement explodes—LiH at* 63.5% R_{eq} *error with* $l_{\max} = 2$ *genuinely insufficient;*
4. *into* ***device-measured many-body overlaps*** *(NOCI): succeeds at the state level, but the operator is dense independently;*
5. *into a* ***permutation symmetry*** *(equivalent-atom swap): net loss, measured—see Observation 3.*

The invariant is that the cost measures the physical asymmetry of a two-center problem. The currency is a choice; the total is not.

The fifth entry was measured for this paper and is the sharpest instance, because the trade is visible term by term.

Observation 3 (Permutation symmetry costs qubits on this builder). *Enabling equivalent-atom-swap tapering* reduces *the qubit saving and inflates the Pauli count, in every configuration tested (Table II). The mechanism is visible in the retained stabilizer labels: the swap rotation merges the per-block Hopf and ℓ-parity stabilizers into joint `SWAP_JOINT` stabilizers, so one permutation stabilizer is bought at the price of several per-block abelian gradings.* ***[MEASURED]***

This is consistent with Theorem 1 rather than incidental to it. A permutation is not an abelian grading of the existing block structure; it is an operation that *identifies* blocks, and identifying blocks destroys the per-block labels that generated the sparsity. The abelian residue is not merely what survives—it is what can be exploited without collapsing the structure that carries it.

Documentation discrepancy. The docstring of `extended_tapered_from_spec` states that atom-swap "saves additional qubits on polyatomics (BeH_2 +1, H_2O +1, NH_3 +1, CH_4 +2)." The measured BeH_2 value is −1 against the Hopf-only baseline and −3 against Hopf + ℓ; the sign is wrong under both. We report the measurement and do not attempt to adjudicate whether the documentation was always incorrect or the behavior regressed.

Observation 2 is labeled an observation, not a theorem: it is a pattern across four instances with a plausible cause, not a proof that no fifth currency exists. The underlying obstruction has been characterized in the framework as non-commuting center projections, with $\|[P_A, P_B]\| = 0.50$ at LiH equilibrium and principal angles 7.6°/44.7°/67.3°.

V. DEMONSTRATION: NAH BINDS WITH GENUINE INTEGRALS

If the wall is the Hamiltonian specification, then supplying genuine end-to-end integrals should bind NaH without any change to the treatment of correlation. It does.

The construction is all-electron 12-electron NaH with Na$\{1s, 2s, 2p\times 3, 3s\}$ and H$\{1s\}$ ($M = 7$ spatial orbitals), non-orthogonal determinants built from fragment-native orbitals, and general-N non-orthogonal Slater–Condon evaluation by Löwdin cofactors. Orbital exponents were optimized on the *isolated* Na atom only ($E(\mathrm{Na}) = -161.0845$ Ha against a Clementi minimal-STO reference of ≈ -161.12), with hydrogen at $\zeta = 1$, so the molecular curve is a fragment prediction rather than a fit. Machinery validation: rotational invariance at 1.1×10^{-13}, and the 91-determinant complete space reproduces bitstring FCI with a span identity of exactly zero.

Three determinants give an interior minimum at +4.8% of the experimental bond length, with the in-basis FCI landing at +0.8%. The D_e deficit (55% of experiment) is minimal-basis incompleteness—no polarization, no diffuse functions—and reproduces the pattern of the four-electron LiH case at 53%; it is not an integral-fidelity effect.

TABLE II. Equivalent-atom-swap tapering against two baselines. ΔQ is qubits removed (larger is better); Pauli counts are for the tapered operator. LiH has no equivalent atoms and is the null control.

system	baseline	ΔQ	ΔQ +swap	Pauli ratio
BeH_2	Hopf only	7	6	3.50×
BeH_2	Hopf + ℓ	12	9	3.60×
N_2	Hopf + ℓ	22	19	2.30×
F_2	Hopf only	12	9	3.50×
F_2	Hopf + ℓ	22	15	3.60×
LiH	Hopf + ℓ	8	8	1.00×

The Löwdin comparators are the informative rows. Orthogonalizing the orbitals and truncating to the *same* number of determinants retains 11.6% of the binding, and the two-determinant case is destroyed outright. Because the complete space is bit-identically the same either way, the damage is specific to *truncated* spaces. The four-electron LiH case replicates this at 15.0% and destroyed, respectively.

Observation 4 (Orthogonalization damages only truncated spaces)**.** *Orthogonalization is span-preserving and therefore harmless on the complete space, and severely harmful on truncated subspaces built from physically meaningful fragment structures. A qubit register only ever represents a truncated space.*

Observation 4 holds at four and twelve electrons, first and second row, with and without p orbitals. It generalizes to systems not yet run and is cheap to falsify.

The mechanism is also visible at *fixed* geometry, without reference to a dissociation limit, which makes it the cleanest form to check. On LiH at $R = 3.25\,a_0$ in a three-orbital fitted-Slater basis, against a complete-space FCI of −8.887049 Ha: three fragment-native determinants land +1.32 mHa above it, while the *same three* determinants built on Löwdin-orthogonalized orbitals land +40.52 mHa above—a 30.6× larger error at identical determinant count. At two determinants the errors are +19.4 mHa and +118.7 mHa (6.1×). The complete spaces agree to machine precision, so the damage is attributable to truncation and not to the orthogonalization corrupting the integrals.

VI. SCOPE

What this paper does not claim. It does not claim the framework's atomic-sector results are affected: single-center angular structure is untouched, exact, and remains the basis of the atomic qubit results. It does not claim the qubit-resource results of Paper 14 are wrong; it clarifies that they are measured on the builder, whose neglected overlap is the subject here. And it does not claim a route around the obstruction, because the census closed the last one available.

A distinction worth naming explicitly, because it bounds any scaling ambition. *Symmetry sparsity*—exact zeros forced by conservation laws, knowable from labels—is what this framework has, and Theorem 1 shows it is an atomic-sector property that degrades to an abelian residue at two centers and to a finite grading when linearity is lost. *Distance sparsity*—approximate near-zeros from spatial locality, exploited by screening, density fitting, and density-matrix truncation—is what linear-scaling electronic structure runs on, and it *improves* with system size. The framework has the first and none of the second. It is, structurally, a small-system framework, and both of its advantages contract as atom count grows: the abelian grading shrinks with symmetry, while the dense cross-center classes grow with the number of center pairs.

TABLE III. NaH, all-electron 12e, $M = 7$. Percentages are of in-basis FCI binding. Experiment: $R_{\mathrm{eq}} = 3.566\,a_0$, $D_e = 1.961$ eV.

method	R_{eq} (a_0)	D_e (eV)	% FCI
covalent (2 det)	3.954	0.686	58.4
$+\,\mathrm{Na}^{+}\mathrm{H}^{-}$ (3 det)	**3.736**	**1.071**	**91.1**
all 4 det	3.713	1.080	91.9
FCI (91 det)	3.595	1.175	100
Löwdin covalent (2 det)	—	unbound	−38.0
Löwdin 3 det	3.658	0.136	11.6

TABLE IV. Claim → backing. LIVE = passing test in `tests/test_paper58_abelian_residue.py`. OWED = result stands on an exploratory driver and must be migrated self-contained into `tests/` before this paper is a finished artifact. CORROB. = numerically corroborated but not exactly decided; see Sec. VI.

claim	tier	backing
Table I S, h	MEASURED	LIVE
Table I g	COUNTED	CORROB.
axial invariance	MEASURED	LIVE
Eq. (1)	SYMBOLIC	LIVE
Thm. 1(i)	INT.	LIVE
Thm. 1(ii)	INT.	LIVE
2-center machinery	SYMB.	LIVE
Obs. 1	MEASURED	LIVE
Pred. 1	UNTESTABLE	(needs angular build)
Obs. 2	OBSERVATION	CHANGELOG v4.73.x
Obs. 3	MEASURED	LIVE
Table III	PANEL-VER.	OWED
Obs. 4	MEASURED	LIVE
fit quality, span id.	MEASURED	LIVE

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